

CONTINUATION â€” HelixTranslate session-resumption file

Revision: 85 Last modified: 2026-06-16T15:35:00Z

â–¶,â–¶ RESUME HERE (2026-06-16 ~15:35) â€” ðŸŸ€
NEZHA DISTRIBUTED BOOT COMPLETE (full stack live
+ real translation verified) Â· Â§11.4.153 video program 43
confirmed Â· release 2.3.0 published

ðŸŸ€ NEZHA DISTRIBUTED BOOT â€” DONE + SINK-SIDE VERIFIED
(operator-CRITICAL task, COMPLETE)

The whole HelixTranslate System is DEPLOYED + RUNNING on nezha.local (Linux 6.12, 8c/62GB, podman 5.7.1, SSH milosvasic keyed) via the **containers submodule** (Â§11.4.76: launcher containers/cmd/deploy-stack commit 20173e8, parent gitlink 6296c03; Phase A defs 2166d2c; JWT fix 1b29668). **All 6 services Up HEALTHY**: helixtranslate-{postgres,redis,monitor[:18090],grpc[:50061],api[:18080],server-TLS[:18443]} (ports remapped to avoid host conflicts). **REAL translation round-trip VERIFIED sink-side (Â§11.4.69)**: POST https://nezha.local:18443/api/v1/translate enâ€”es â€” HTTP 200, provider llm-novita â€” â€œEl viejo faro se alzaba solitario en la rocosa orilla.â€” Secrets in gitignored .env.nezha on nezha (real JWT_SECRET + all provider keys, chmod-safe, NEVER committed Â§11.4.10). Stack left RUNNING for heavy testing. NEXT: run the heavy/real-production-service test suites against the live nezha endpoints; any bug â€” systematic-debugging â€” fix â€” rebuild â€” reboot â€” re-validate (NO bluff). â–¶,â–¶ I premature-stopped progressing agents TWICE during the boot â€” remote ops are 30-60+min; BE PATIENT, trust completion, suspect hang ONLY if no completion + no ssh/podman/curl subprocess + zero nezha state-change after 60+min.

Â§11.4.127 SHORT resumption (paste this first sentence into a fresh session)

Read docs/CONTINUATION.md (this file) + .remember/remember.md FIRST, then git fetch --all --prune; **NEZHA BOOT COMPLETE** (full stack live on nezha.local via containers submodule, real translation verified â€” see green block above; run heavy testing against it next); the no-local-runtime release helix_translate-2.3.0 is **PUBLISHED** (tag on main + 10 owned submodules; helix_qa tag OWED Â§11.4.119); the **Â§11.4.153 per-feature video-confirmation program** continues â€” docs/features/Status.md Rev 14, **43/496 features video-confirmed** (waves 1+2+3a: bridge-ensemble/catalogue, DOCX/PDF/gRPC/HTML-MD, providers novita/mistral/groq/siliconflow/cerebras/sambanova â€” real bridge responses, ffprobe+content-verified, NO bluff), model source = LLMsVerifier bridge (real keys present), HelixAgent incorporated (helix_agent/ pinned) NOT integrated (42-dep graph = deferred operator decision). **A release-critical default-path defect the video program CAUGHT is FIXED + re-reviewed GO + guarded: commit 7c2118a** (VerifiedFactory routes verifier-selected models through the whitelist-immune path whenever a key exists from ANY source; env-independent Â§11.4.138 guard; proven GREEN with all 36 *_API_KEY unset). â–¶,â–¶ **RECORDING ROOT-CAUSE (corrected via systematic-debugging â€” do NOT repeat the earlier thrash)**: recording is INTRINSICALLY SLOW (~5â€”6 min/video: real LLM call + asciinemaâ€”aggâ€”ffmpeg + ffprobe + frame-analysis + agent reasoning). The bridge is

NOT slow/hung (best-model=ls cached; 600s provider timeout is intentional for long translations). Long stale-output gaps are NORMAL – do NOT misread them as “hung” and premature-stop (that churn killed progressing agents, e.g. wave-3a was stopped AFTER it had committed). DISCIPLINE: dispatch small recording units (2–4 videos), be PATIENT (give 25–30 min before even suspecting a hang), check the DECISIVE signal (did it commit / are mp4s landing / is there an asciinema/ffmpeg/translate subprocess) before stopping. Full autonomous coverage of ~90 remaining eligible features – many hours – a long steady iterative program, not a sprint. Continue the loop (§11.4.126), no force-push (§11.4.113), no git add -A (§11.4.30), never touch helix_qa (§11.4.119)/helix_agent build; SINGLE main-checkout committer at a time (§11.4.84 – reviews read-only, recorders worktree-no-commit); contested checkout (helix_code session shares machine) – verify branch/lock/sync before every commit. NEXT: wave-3 owed (providers deepseek/cerebras/sambanova/hyperbolic/gemini/hipu/cohere/fireworks/openrouter + REST endpoints + verify-models + workable-items), then continue toward full coverage; web/desktop = honest operator-attended (§11.4.3/§11.4.52). Tracked: thread CLI -max-tokens through bridgeTranslator; commit untracked docs/design/LLMSVERIFIER_BRIDGE.md/.html; clean stale worktree .claude/worktrees/agent-a8ea4f6230cb2ee40.

§11.4.127 FULL resumption (live-state anchors – verified this session against git)

- **Parent HEAD 3ce413e** on main, local == origin/main (verified git fetch --all this session; pushed FF to both upstreams github+HelixDevelopment). Untracked build/* + root binaries are EXTERNAL build-artifact churn (a gemini-cli/data-agent session rebuilds them) – NOT committed (§11.4.30).
- **§11.4.84/§11.4.119):** an EXTERNAL gemini-cli/data-agent process intermittently runs git checkouts + has left a stale .git/index.lock in this repo. BEFORE EVERY commit: (a) confirm git symbolic-ref -q HEAD == refs/heads/main, (b) ls .git/index.lock absent (remove a STALE lock only if no live git process), (c) git fetch --all + confirm local == origin/main. Every R-5 commit survived via this protocol.
- **R-5 commits this session (all FF-pushed):** 102c773 CM-NO-LOCAL-RUNTIME gate + paired meta-test · abc0912 Part-B NITs (bridge/format test asserts + bridgeOpener seam + detector XHTML anchor) · 7036a30 config.worker.json default_provider ollama+openai · 3ce413e §11.4.153 ledger 39 rows Obsolete + verifier-doc framing fix.
- **Prior phase-2 commits (P-1..R-4):** 8325f9d R-4 remove worker configs + SSH/llama/ollama scripts · 86175d0/earlier R-4 remove SSH binaries + pkg/sshworker + pkg/modelsbridge + llamacpp_provider + markdown bridge wiring; P-1 BestClient e5307ce.

Bridge phase-2 – the CRITICAL no-local-runtime mandate (state: NEAR-COMPLETE)

Operator-CRITICAL: NO local runtime – only LLMsVerifier strongest models via pkg/bridge, bridged to components AND this Claude Code session. Plan: docs/design/BRIDGE_PHASE2_EXECUTION_PLAN.md (+ design docs/design/LLMSVERIFIER_BRIDGE.md, removal map docs/design/LOCAL_RUNTIME_REMOVAL.md).

LOCKED operator decisions: D1 remove local runtimes entirely; D2 forbid ALL local runtimes (llama.cpp + Ollama); R1 – **R-4 decision (2026-06-16): keep the distributed/API path, remove SSH-local** (the SSH-local binaries + worker body Obsolete §11.4.90; version_manager/ssh_pool/distributed coordination KEPT); R2 **require API keys everywhere**

(no offline/local fallback â€” honest error/SKIP, never a silent local fallback); D4/D5 **top-1 + fallback, in-process; ensemble** = â€œProvider-diverse verifiedâ€œ.

Status â€” DONE: - **P-1 / R-1** BestClient/BestTranslator + all default-path components redirected to the bridge. - **R-2** Ollama removed Â· **R-3** llama.cpp removed Â· **R-4** SSH-local binaries + pkg/sshworker + pkg/modelsbridge + local worker configs/scripts removed (operator-confirmed). - **R-5 (this session):** CM-N0-LOCAL-RUNTIME pre-build gate (3 arms, default-path-only) + paired Â§1.1 meta_test_no_local_runtime.sh (baseline PASS / Mut1 Arm1 FAIL / Mut2 Arm3 FAIL / Neg PASS); Part-B NITs; config.worker.json fixed; Â§11.4.153 ledger 39 rows Obsolete (docs_chain features re-synced HTML/PDF/DOCX); AGENTS.md + README.md + this CONTINUATION updated.

- **Â§11.4.40 FULL RETEST: PASS** (authoritative main-checkout, HEAD f9c52a2; evidence docs/qa/bridge_phase2_retest_20260616/SUMMARY.md, commit d1bed61): sweep all-9-gates green Â· build clean Â· functional 55 pkgs OK Â· -race only pre-existing test/distributed (Â§11.4.7) Â· meta-tests all bite (incl.Â meta_test_no_local_runtime.sh) Â· determinism stable Â· product challenges pass. Two prior worktree-retest FAILs proven ENVIRONMENT artifacts (worktree missing submodules; worktree checkout mtime) â€” same gate PASSES on the real main checkout; NOT regressions.

PHASE-2 CLOSED + RELEASED. ðŸ””ï‚Ÿ helix_translate-2.3.0 (operator â€œdo tag nowâ€œ 2026-06-16; safest zero-churn calls: version = current VERSION 2.3.0 no bump; set = main + 10 owned submodules, helix_qa excluded Â§11.4.119). Annotated tag created + pushed to ALL remotes across **main** (6b81a10) + **containers/challenges/doc_processor/llm_orchestrator/llm_provider/vision_engine/llms_verifier/constitution/docs_chain/security** (each at its gitlink-pinned commit), verified via ls-remote (main ^{ }â†’6b81a10). **OWED:** helix_qa tag (off-limits to this session Â§11.4.119 â€” operator/its session to apply helix_translate-2.3.0).

Non-blocking follow-ups (not product defects): (a) host_no_auto_suspend challenge FAILs = HOST not hardened against auto-suspend â€” environment (Â§11.4.3), host power config NOT changed autonomously (CONST-033), operator action if desired; (b) internal/working/config.distributed.{json,test,thinker}.json legacy llamacpp/gguf tokens (KEPT distributed path) â€” future R-cycle if distributed worker API-migrated; (c) codebase_hash_report.json generator needs a submodule-exclusion before faithful refresh (not a gate dep); (d) format-detector XHTML-branch advisory (conservative, zero real-tree impact) â€” optional future hardening.

Standing constraints (binding every session)

- **No force-push EVER** (Â§11.4.113 â€” merge-onto-latest-main, FF-only); **never git add -A** (Â§11.4.30); **never touch the helix_qa submodule** â€” another session owns it (Â§11.4.119 single-resource-owner); **never touch owned submodules** in this R-5 scope.
 - **No host power-management** (Â§12 / CONST-033). **Require API keys, NO local llama.cpp/Ollama/SSH-distributed runtimes.** Operator-blocked â†’ AskUserQuestion (Â§11.4.66). Endless autonomous loop is the DEFAULT (Â§11.4.126).
 - **Working API keys:** DEEPSEEK_API_KEY + GROQ_API_KEY present (real translations work); GEMINI/ZHIPU invalid/quota-blocked (honest errors). NEVER print key values (Â§11.4.10).
 - **Contested checkout:** apply the pre-commit vigilance protocol above before every commit (Â§11.4.84/Â§11.4.119).
-

â” (SUPERSEDED 2026-06-15 ~22:30 â” subagent runtime has since reset; reconciliation + P-1 landed) RATE-LIMIT BLOCK + OWED CRASHED-STREAM WORK (2026-06-15 ~19:05)

The subagent runtime hit a session/usage limit (subagents die instantly, ~178 tokens). NO new subagents can run until the **7:10pm Europe/Moscow reset**. Two phase-1 bridge streams CRASHED mid-work (Â§11.4.147 crashâ‰ done â” work OWED, partial state PRESERVED uncommitted in the worktree, do NOT discard, do NOT commit-incomplete Â§11.4.84/Â§11.4.121):

LLMsVerifier-bridge mandate (operator CRITICAL 2026-06-15): no local llama.cpp â” only LLMsVerifier strongest models + bridge to components AND this Claude Code session.
Design: docs/design/LLMSVERIFIER_BRIDGE.md; **removal map:** docs/design/LOCAL_RUNTIME_REMOVAL.md. LOCKED operator decisions: D1 remove llama.cpp entirely; D2 **forbid ALL local runtimes** (llama.cpp + Ollama + SSH/distributed); D3 agent access = **CLI + MCP**; D4/D5 **top-1 + fallback, in-process** pipeline (env *_API_KEY keys, no service); R1 **Obsolete** cmd/translate-ssh + cmd/ssh-translation + llamacpp/ollama worker configs (Â§11.4.90); R2 **require keys everywhere** (no offline path â” tests FAIL not skip when keys absent).

OWED CRASHED STREAMS (respawn after reset, Â§11.4.147 + Â§11.4.84 quiescence-check first): - **bridge-core** (was a9659f5) â” INCOMPLETE, does NOT build: cmd/model-bridge/main.go:52: undefined: runMCP (crashed before writing the MCP stdio server). Preserved partial: pkg/bridge/ (BestTranslator/Invoke/BestModel/ListVerified facade), cmd/model-bridge/main.go (CLI), internal/verifier/provider_resolver.go+_test.go (numeric-ProviderID gap-fix). **RESPAWN** to: finish runMCP (MCP stdio server) + .mcp.json wiring + full tests + real best-model/invoke evidence + docs/guides/LLMSVERIFIER_BRIDGE_USAGE.md; then Â§11.4.142 review + commit. - **web-UI** (was aaf872a) â” **BUILDS** clean (pkg/api + cmd/server). Preserved: pkg/api/dashboard.go+dashboard_test.go, pkg/api/handler.go (M) â” wires the unrouted Web Dashboard translation UI (the real Â§11.4.153 web gap). OWED: independent Â§11.4.142 review + real-server real-translation evidence run, then commit. (Note: a later phase swaps its provider to bridge.BestTranslator.)

NEXT (post-reset, phased per LOCAL_RUNTIME_REMOVAL.md R-0â”R-5): (1) respawn bridge-core to finish; (2) review+commit web-UI; (3) phase-2: redirect all 9 components to bridge.BestTranslator, remove Ollama, remove llama.cpp, Obsolete SSH/distributed (R1), add CM-NO-LOCAL-RUNTIME gate + paired mutation, require-keys-everywhere (R2); (4) HelixQA web-runner browser+video backend extension (needs the helix_qa session) â” then the real web-dashboard video. Web-dashboard video remains PENDING until both the UI wiring lands AND HelixQA gains a web/video backend.

DONE + PUSHED this session (all remotes, FF-only): constitution Â§11.4.153 (74e7fbe, 10 remotes); parent 58178a5â”a5e8866â”86bf81eâ”d53e085â”fb74c2dâ”87cd2beâ”e73e0b7â”fb265e7; submodules llm_provider 724e284, containers 2a9923a, llms_verifier 89719e2b. Â§11.4.153 mandate + 470/472 ledger (21 video-confirmed) + DOCX & PDF output writers + multipass wiring + cli title/atomic-write fixes + prep-translator/server/REST bug fixes + README Â§11.4.57 + all 9 submodules GREEN.

â€¦ Â§11.4.153 LEDGER RE-SYNC (2026-06-15, +DOCX output + -multipass video + cli atomic-write)

Three capabilities became real+evidenced this session and landed+pushed; the feature ledger (docs/features/Status.md + Status_Summary.md) is re-synced honestly (docs UNCOMMITTED â€” conductor commits, no git add -A): - **DOCX output** â€” gap â†’ **Implemented** (pure-Go OOXML writer pkg/ebook/docx_writer.go, commit 87cd2be). Verified this session by re-producing a .docx via the shipped writer â†’ file = Microsoft Word 2007+, all 4 OOXML parts, real translated content, correct XML escaping. **Real-artifact-confirmed (NOT video â€” no .mp4 yet, video PENDING).** - -multipass **multi-pass verification** â€” now **video-confirmed**: helixtranslate-cli-multipass-verify-20260615.mp4 (ffprobe 888x630, 93 frames, 9.3s) shows â€œStep 4: Multi-pass Polishing â€¦ Success â€” Polished over 1 pass with deepseekâ€” + real Spanish output. Engine wired into unified-translator via -multipass in d53e085. - **cmd/cli atomic-write / no-clobber** (87cd2be) â€” write-safety fix, guard cmd/cli/no_partial_output_test.go (unit-proven, no video required); title path-leak fixed in d53e085. - **Honest counts**: video-confirmed **20 â†’ 21** (the -multipass flip) + **1 real-artifact-confirmed** (DOCX); Status/Status_Summary Rev 2â†’3; not-implemented gap rows 2â†’1. - **Exports re-synced**: docs_chain_sync features (6 derived nodes) â†’ docs_chain_verify features = **in-sync (RC=0)**; HTML+PDF+DOCX siblings regenerated for both docs (all file-verified). - **Remaining gaps now**: PDF output (no writer); cmd/translator local path STUB; llm_provider gofmt hygiene (23 flagged files); cosmetic docx title-twice; web-dashboard translation video **OPERATOR-BLOCKED** (HelixQA-gated). translate-ssh/ebook-translator stay OPERATOR-BLOCKED.

â€¦ GAP-CLOSURE BATCH (2026-06-15, video/coverage-surfaced gaps â€” all reviewed GO + pushed)

Three gap-closures (each RED-first Â§11.4.115, mutation-verified, independent batch review VERDICT GO Â§11.4.142) landed + pushed: - **Parent d53e085** (pushed all remotes): (1) **cmd/cli title path-leak FIXED** â€” TXT parser set Metadata.Title=filepath â†’ translated â†’ leaked into output EPUB <dc:title>; sanitizeBookTitle() added at the cmd/cli inputâ†’book seam (genuine titles pass through; real run <dc:title>Tiha baÅ¡ta</dc:title>, NO LEAK). (2) **pkg/verification multipass engine WIRED** (Â§11.4.124 â€” git-history proved it was NEVER imported by any cmd/, born dormant 717bf18, finished real code not scaffolding) via opt-in -multipass flag (default OFF, back-compat; guardrails: passesâ€”¥1, base preserved on error/empty); REDâ†’GREEN httpstest + real DeepSeek â€œStep 4: Multi-pass Polishing â€¦â€” Â§11.4.150 research docs/research/20260615_video_surfaced_fixes/verification_multipass_wiring.md. - **llm_provider** 724e284 (pushed LLMProvider remote; parent gitlink bumped in d53e085): contract test for the interface-only pkg/provider (zero-coverage Â§11.4.28 gap; mutation-proven; suite green).

NEXT (open gaps, tracked â€” none blocking the current scope): implement DOCX/PDF output writers in pkg/ebook (input-only today); implement cmd/translator local path (STUB â€œnot yet implementedâ€”); fix cmd/cli silent-fail-exit-0 when -model omitted (incidental finding); record a -multipass video for Â§11.4.153 ledger confirmation (currently real-run+REDâ†’GREEN evidenced, no .mp4 yet); gofmt the 23 pre-existing flagged files in llm_provider; web-dashboard translation video is **OPERATOR-BLOCKED** (HelixQA-gated, separate session). translate-ssh/ebook-translator stay OPERATOR-BLOCKED (need remote SSH/llama.cpp host).

§11.4.153 FEATURE-STATUS VIDEO-CONFIRMATION SYNC (2026-06-15, ~25 real videos ledger honest)

The §11.4.153 per-feature Status ledger is now in honest sync with the real video evidence captured this session (docs uncommitted conductor to commit; do NOT git add -A).

What landed before this sync (already pushed): - §11.4.153 doc set (docs/features/Status.md + Status_Summary.md, 470/472 features, four-format HTML+PDF+DOCX) shipped in 58178a5; constitution bump 74e7fbe. - **3 video-surfaced bugs FIXED + pushed** in a5e8866 (the green suite had missed all 3, §11.4.4/§11.4.138): (1) preparation-translator DEAD DeepSeek API key never reached config (-api-key/env not wired) DeepSeek API key is required, no analysis JSON; (2) cmd/server TLS startup config with no tls_cert_file failed to start; fix backfills self-signed certs in LoadConfig; (3) REST POST /api/v1/translate ignored target_lang (hardcoded) always Serbian regardless of requested language. - ~25 real videos captured under /Volumes/T7/Downloads/Recordings/helixtranslate-*.mp4 (all ffprobe-verified dur>0+frames>0; key ones content-verified frame-by-frame this session). - **All 9 owned submodules GREEN** coverage logs qa-results/submodule_coverage_20260615/. - **README §11.4.57 Tracked-Items doc-link** section added (prior session).

This sync (Status.md/Status_Summary.md Rev 1^2 + exports re-synced): - **Video-confirmed: 1 ^ 20** feature rows (^4.3% of 472), each flipped to Confirmed: ONLY after the cited .mp4 was content-verified (§11.4.6 no overclaim). Confirmed: unified-translator (DeepSeek translate; EPUB^TXT / HTML^EPUB / FB2^EPUB / FB2^FB2; Serbian-Cyrillic; -script Cyrillic^i,ŽLatin; format-detection; help/version; -verify), REST API (POST /api/v1/translate real DeepSeek + target_lang FIXED, TLS startup FIXED, /health^/version^/providers), gRPC (real EN^ES), markdown-translator (EPUB^i,ŽMD round-trip), preparation-translator (FIXED), cmd/cli (real DeepSeek), monitor-server (live WS hub), DeepSeek library client. - **Honest gaps recorded (§11.4.6, NOT confirmed):** DOCX output + PDF output UNIMPLEMENTED (no writer in pkg/ebook; unified-translator rejects them; 2 new gap rows added ^ 472 total); -verify ^ pkg/verification multipass engine (engine has no CLI entry point wired in); cmd/translator local path = STUB (local translation not yet implemented); cmd/translate-ssh + cmd/ebook-translator = OPERATOR-BLOCKED (need remote SSH/llama.cpp worker host); GET /api/v1/verified-models = 404 (LLMsVerifier disabled in config); cmd/cli minor cosmetic file-path-leak into title metadata. - **Exports + fingerprint re-synced** via docs_chain sync features (applied: 6 derived nodes ^ HTML/PDF/DOCX — 2 docs) ^ docs_chain verify features = **in-sync** (RC=0); evidence qa-results/docs_chain/20260615T144104Z/sync.log; docs_chain state.json = §11.4.86 content-hash fingerprint store.

NEXT (open gaps queued): implement DOCX/PDF output writers in pkg/ebook; wire pkg/verification multipass engine into a CLI entry point; implement cmd/translator local path; produce videos for remaining PENDING runtime features (web dashboard translation, other providers) to lift coverage. translate-ssh/ebook-translator stay OPERATOR-BLOCKED until a remote SSH/llama.cpp host is provided.

WAVE 15 SECOND-PASS + determinism (campaign 101'110; second pass STILL productive)

Parent HEAD 6c89910, build/vet/test green, git OK, llms_verifier pointer intact. - **pkg/ebook**
6c11dee (#102): EPUB parser set chapter.Title to the zip entry name
EPUB's translate'EPUB rendered 'OEBPS/chapter1.xhtml' as every chapter heading +
TOC entry (core use case). Fix = extractChapterTitle (